

Extended Kalman Filter based Robot Localization Using a Ray-Casting Sensor Model

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Abstract—We present an Extended Kalman Filter (EKF) for localizing a TurtleBot3 on a pre-built occupancy grid map under ROS 2. Wheel odometry drives the predict step, while a ray-casting sensor model replaces classical landmark extraction in the update step and reuses the same map representation to score globally distributed pose hypotheses for kidnapping recovery. The system is evaluated in three structurally different environments and shown to track the robot reliably under external disturbances as well as to re-localize successfully after manual displacement, with localization accuracy below pure odometry in all scenarios.

I. INTRODUCTION

Reliable self-localization on a known map is a prerequisite for almost any autonomous mobile robot. Wheel odometry is locally accurate but accumulates drift, while a single LIDAR scan, although globally informative, is noisy, sparse, and ambiguous in repetitive environments. The Extended Kalman Filter (EKF) addresses this by fusing both sources into a single Gaussian belief over the pose, weighting them by their respective uncertainties. Its predict–update structure keeps the computational cost low enough for online operation on embedded hardware, a natural fit for small differential-drive platforms.

In this work we implement and evaluate such an EKF-based localizer on the TurtleBot3 running under ROS 2. The robot drives autonomously on a pre-built occupancy grid map and the filter runs online, fusing the on-board wheel odometry with the 2D LIDAR scans provided by the platform. Instead of the classical pipeline that extracts geometric features from the scan and matches them against landmarks stored in the map, we use a ray-casting sensor model: for each candidate pose, expected ranges are obtained by marching beams through the occupancy grid until an occupied cell is hit. This avoids both the feature-extraction step, sensitive to noise from thin obstacles, and the data-association step, ambiguous in repetitive environments.

Beyond the standard predict–update cycle, the filter is also used as the backbone for a kidnapping–recovery mechanism. The same ray-casting model that drives the EKF update is reused to score globally distributed pose hypotheses against the live scan whenever filter health degrades, so the robot can re-localize after being manually displaced without restarting the node. The remainder of the report describes the EKF algorithm (Sec. II), the implementation details around the ray-casting model and the kidnapping recovery (Sec. III), and

an experimental evaluation across three structurally different environments (Sec. IV).

II. EKF ALGORITHM

The Extended Kalman Filter (EKF) maintains a Gaussian belief over the robot pose and iterates between a *predict* step, driven by odometry, and an *update* step, driven by LIDAR-landmark observations (Fig. 1). The formulation used in this work follows the practical sensor-fusion variant for mobile-robot localization of [1].

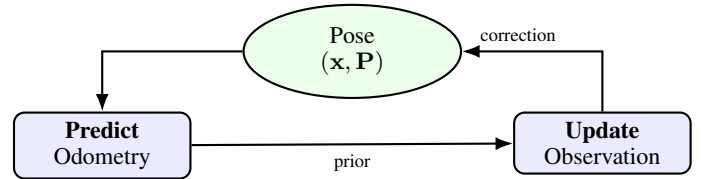


Fig. 1. EKF predict–update cycle.

A. State Representation

The filter state is the planar pose $\mathbf{x} = [x \ y \ \theta]^T$, and estimation uncertainty is encoded in the 3×3 covariance matrix

$$\mathbf{P} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{x\theta} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{y\theta} \\ \sigma_{\theta x} & \sigma_{\theta y} & \sigma_{\theta\theta} \end{bmatrix}, \quad (1)$$

whose diagonal entries are variances and off-diagonal entries are cross-correlations.

B. Predict Step

Odometry provides the linear velocity v and angular velocity ω , which integrate over time step T into the control input $\mathbf{u} = [d, d_\theta]^T$ with $d = vT$ and $d_\theta = \omega T$. The nonlinear motion model is

$$\mathbf{x}' = g(\mathbf{x}, \mathbf{u}) = \begin{bmatrix} x + d \cos(\theta + d_\theta) \\ y + d \sin(\theta + d_\theta) \\ \theta + d_\theta \end{bmatrix}. \quad (2)$$

Linearizing g around the current estimate yields the state Jacobian $\mathbf{F}$ and control Jacobian $\mathbf{G}$:

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & -d \sin(\theta + d_\theta) \\ 0 & 1 & d \cos(\theta + d_\theta) \\ 0 & 0 & 1 \end{bmatrix}, \quad (3)$$

$$\mathbf{G} = \begin{bmatrix} \cos(\theta + d_\theta) & -d \sin(\theta + d_\theta) \\ \sin(\theta + d_\theta) & d \cos(\theta + d_\theta) \\ 0 & 1 \end{bmatrix}. \quad (4)$$

The predicted covariance is

$$\mathbf{P}' = \mathbf{F} \mathbf{P} \mathbf{F}^T + \mathbf{G} \mathbf{Q}_k \mathbf{G}^T, \quad (5)$$

where the diagonal process noise matrix $\mathbf{Q}_k = \text{diag}(\sigma_d^2, \sigma_\theta^2)$ models translational and rotational actuation uncertainty.

C. Update Step

Whenever a sensor measurement $\mathbf{z}$ is available, the filter corrects the prior belief. A nonlinear observation model $h(\mathbf{x})$ predicts the expected measurement from the current state estimate. Its linearization $\mathbf{H} = \partial h / \partial \mathbf{x}$ maps state space into measurement space. The concrete form of h depends on the sensor and is detailed in Section III.

The innovation $\mathbf{y} = \mathbf{z} - h(\hat{\mathbf{x}})$ captures the mismatch between observation and prediction. The Kalman gain

$$\mathbf{K} = \mathbf{P} \mathbf{H}^T (\mathbf{H} \mathbf{P} \mathbf{H}^T + \mathbf{R}_k)^{-1} \quad (6)$$

balances trust in the prior against trust in the measurement. The corrected state and covariance are

$$\hat{\mathbf{x}} \leftarrow \hat{\mathbf{x}} + \mathbf{K} \mathbf{y}, \quad \mathbf{P} \leftarrow (\mathbf{I} - \mathbf{K} \mathbf{H}) \mathbf{P}. \quad (7)$$

D. Tunable Parameters

The two noise matrices are the primary design knobs. Process noise $\mathbf{Q}_k$ encodes trust in the motion model. Larger values of σ_d^2 or σ_θ^2 cause $\mathbf{P}'$ to grow faster, making the filter more receptive to incoming measurements. Setting $\mathbf{Q}_k$ too low leads to odometry drift accumulation. Too high makes the filter over-reactive. Measurement noise $\mathbf{R}_k$ encodes sensor precision. A smaller $\mathbf{R}_k$ gives measurements more weight relative to the prior, while a larger $\mathbf{R}_k$ discounts them. In practice it is the ratio $\mathbf{Q}_k / \mathbf{R}_k$ that determines whether the filter leans on odometry or on sensor corrections, and both matrices are tuned jointly against localization error on representative trajectories.

III. IMPLEMENTATION DETAILS

A. From Landmarks to Ray Casting

The classical landmark-based update requires two steps before any EKF correction can be applied: extracting geometric features from the raw scan and matching them to landmarks stored in the map. Both steps introduce failure modes. Feature extraction must be robust against noise from thin obstacles such as chair legs, which appear only intermittently and at varying scan angles. Data association then needs to resolve which extracted feature corresponds to which map landmark, a problem that becomes ambiguous in repetitive environments.

Together, these steps add considerable engineering effort and are a source of localization failures that are difficult to attribute to the filter itself.

Ray casting sidesteps both issues by working directly on the raw LIDAR ranges and the occupancy grid (Fig. 2). For a candidate pose, the expected range along each beam is computed by marching through the map until an occupied cell is hit. No feature extraction is needed, and there is no association problem because every beam votes independently against the same grid.

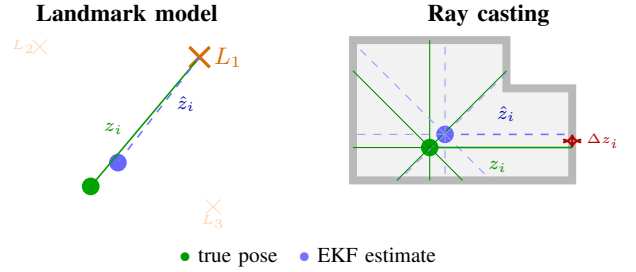


Fig. 2. Landmark model (left) vs. ray-casting model (right).

B. Ray-Casting Measurement Model

The full LIDAR scan is subsampled to $N = 30$ uniformly spaced beams with angles $a_1, \dots, a_N$ relative to the robot heading. For a given pose estimate $\hat{\mathbf{x}} = [x, y, \theta]^T$, each expected range is obtained by marching along the corresponding ray at step size $\Delta s = 0.5$ res until an occupied cell is reached or the maximum range $r_{\max} = 3.5$ m is exceeded:

$$\hat{z}_i = \min\{k\Delta s \mid \text{cell}(x + k\Delta s \cos \alpha_i, y + k\Delta s \sin \alpha_i) = \text{occ}\}, \quad (8)$$

where $\alpha_i = \theta + a_i$ and occ denotes an occupied cell. The full observation function is

$$h(\hat{\mathbf{x}}) = [\hat{z}_1, \dots, \hat{z}_N]^T \in \mathbb{R}^N. \quad (9)$$

Measurement noise is modelled as isotropic Gaussian with $\mathbf{R}_k = \sigma_R^2 \mathbf{I}_N$.

Before the update, beams are filtered through an innovation gate: only beams for which both the observed and predicted ranges are below $r_{\max}$ and the absolute innovation $|z_i - \hat{z}_i| < \delta = 0.157$ m are passed to the Kalman gain computation. This rejects saturated readings and outliers caused by dynamic obstacles not present in the map.

C. Observation Jacobian via Central Differences

Because the ray-casting function has no closed-form derivative with respect to the pose, the Jacobian $\mathbf{H} \in \mathbb{R}^{N \times 3}$ is approximated numerically using central differences with step size $\varepsilon = 0.035$ m:

$$H_{ij} = \frac{h_i(\hat{\mathbf{x}} + \varepsilon \mathbf{e}_j) - h_i(\hat{\mathbf{x}} - \varepsilon \mathbf{e}_j)}{2\varepsilon}, \quad (10)$$

where $\mathbf{e}_j$ is the j -th standard basis vector. This requires six ray-casting evaluations per update step (two per state dimension) on the subset of valid beams only, keeping the computational cost manageable.

D. Kidnapping Detection and Recovery

The node monitors filter health on every scan callback via two criteria: the fraction of beams that pass the innovation gate must exceed a minimum ratio $\rho_{\min} = 0.25$, and the trace of the covariance matrix must remain below $\text{tr}(\mathbf{P}) < 5.0$. If either condition is violated for five consecutive scans (LOST_CONSEC_STEPS), the robot is declared lost and a three-phase recovery sequence begins.

In the first phase (*waiting*), the EKF switches to odometry-only prediction for one second while buffering incoming scan-odometry pairs. This gives the robot time to move slightly, making the subsequent sensor comparison more informative. After the waiting period, the second phase launches a global search in a daemon thread to avoid blocking the scan callback.

The search evaluates hypotheses placed on a regular grid over all free cells of the map. The grid spacing of 0.40 m with $N_\theta = 16$ uniformly distributed orientations was determined by a prior analysis: at this resolution, the highest-scoring candidate lies reliably within the convergence basin of the EKF whenever an accurate initial pose had been provided, confirming that the grid is dense enough not to miss the true location. Each hypothesis is scored by dead-reckoning it through the buffered odometry and comparing the predicted raycasts to the observed scans at every step. Beams with an innovation below the recovery gate of 0.5 m count as valid, and the final score is

$$s = \frac{n_{\text{valid}} - \mu \cdot \bar{\iota} \cdot n_{\text{eval}}}{n_{\text{eval}} \cdot N}, \quad (11)$$

where $\bar{\iota}$ is the mean innovation over valid beams, n_{eval} is the number of evaluated buffer steps, and $\mu = 0.5$ is an innovation-penalty weight. The winning hypothesis is propagated forward through the buffer to obtain the pose at the current instant and tracking resumes with a reset covariance $\sigma_{\text{rec}} = 0.3$ m.

The same mechanism also serves as the initial localization strategy. The node starts directly in global-search mode, so if no `/initialpose` is provided the filter locates the robot autonomously before entering tracking.

E. Node Architecture

The node subscribes to `/odom` (odometry), `/scan` (LIDAR), and `/initialpose` (optional manual pose override). It publishes the map on `/map` with TRANSIENT_LOCAL durability, the `map→odom` TF after every odometry message, and beam visualisations on `/ekf/beams` for RViz. The scan callback drives all filter logic: predict and update during tracking, buffer collection during global localisation, and odometry-only prediction during the waiting phase. Global localisation runs in a separate daemon thread to avoid blocking the scan callback.

IV. EXPERIMENTAL EVALUATION

The localization system was evaluated in three distinct environments, each chosen to stress a different aspect of the

pipeline. A small, geometrically clean room served as the baseline to verify basic correctness. A larger, cluttered lab tested robustness against sensor noise introduced by furniture. Finally, a long corridor with repetitive structure was used to show how much more significant the impact of the noise parameters $\mathbf{Q}_k$ and $\mathbf{R}_k$ becomes when the geometry stays uniform along the direction of travel. All experiments were conducted on a pre-built map, with the robot navigating autonomously while the EKF ran online.

A. Elevator Room — Baseline



Fig. 3. Reference map of the elevator room used as baseline environment.

The elevator room (Fig. 3) is a small, rectangular space with sharp, well-defined walls and no obstacles. Its simple geometry produces highly consistent LIDAR returns, making it an ideal environment to verify that the filter converges and maintains a stable estimate under near-ideal conditions.

The baseline was driven aggressively, with fast translations and tight looped turns, in order to stress the predict step. Around the nominal noise values introduced in Sec. II-D, we performed a sweep on $\mathbf{Q}_k$ and $\mathbf{R}_k$: σ_{QD} and $\sigma_{Q\theta}$ were divided by an integer factor k while σ_R was multiplied by the same factor, yielding configurations EKF $k\times$ with $k = 1, \dots, 6$. Larger k therefore shifts trust from the LIDAR towards odometry. Ground truth was obtained from the ROS 2 `amcl` module running in parallel.

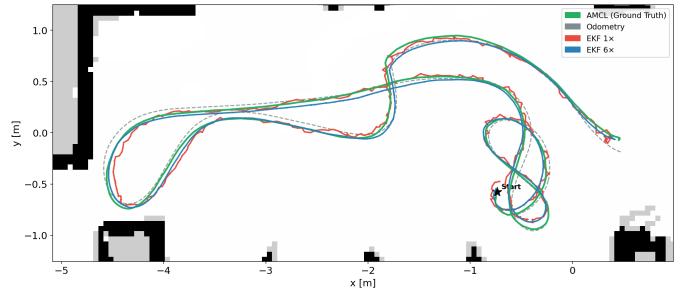


Fig. 4. Trajectories in the undisturbed elevator-room run.

A first observation is that the TurtleBot's odometry is already very accurate in this benign scenario (RMSE 0.066 m, grey dashed trace in Fig. 4). Increasing $k\times$ therefore yields

a visibly smoother estimated trajectory and a monotonically decreasing RMSE that drops from 0.098 m at EKF 1 $\times$ to 0.051 m at EKF 6 $\times$ (Tab. I).

To create a more demanding scenario, the robot was manually disturbed during a second run by pushing and rotating it while it was driving. The EKF behaviour stays qualitatively the same and all configurations remain locked onto the ground truth, whereas the pure-odometry RMSE jumps from 0.066 m to 0.342 m. The visible drift of the grey trace from the green AMCL reference in Fig. 5 makes the necessity of the filter in any realistic deployment explicit.

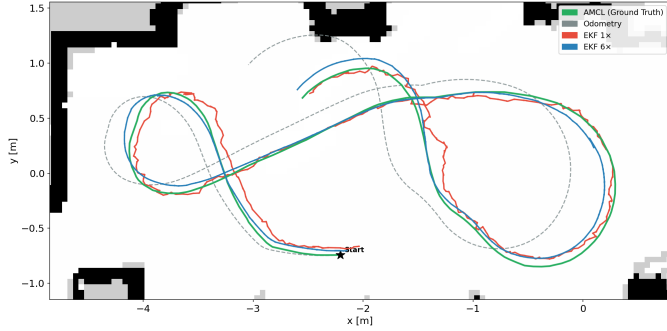


Fig. 5. Trajectories under external disturbances.

Notably, the relative gap between the baseline EKF 1 $\times$ and the odometry-heavy EKF 6 $\times$ narrows under disturbance, which is consistent with the underlying assumption being amplified, namely that odometry is reliable, no longer holding once the robot is pushed externally.

TABLE I
LOCALIZATION RMSE (M) FOR THE UNDISTURBED AND DISTURBED ELEVATOR-ROOM RUN.

Configuration	$(\sigma_{QD}, \sigma_{Q\theta}, \sigma_R)$	Undist.	Dist.
Odometry	—	0.066	0.342
EKF 1 $\times$	(0.0500, 0.090, 0.06)	0.098	0.127
EKF 2 $\times$	(0.0250, 0.045, 0.12)	0.091	0.107
EKF 3 $\times$	(0.0167, 0.030, 0.18)	0.084	0.100
EKF 4 $\times$	(0.0125, 0.022, 0.24)	0.075	0.098
EKF 5 $\times$	(0.0100, 0.018, 0.30)	0.061	0.092
EKF 6 $\times$	(0.0083, 0.015, 0.36)	0.051	0.078

B. Lab Room — Robustness and Start-Pose Sensitivity

The lab (Fig. 6) is a larger room populated with tables, chair legs, and other thin obstacles that are not always detected consistently by the LIDAR. This increases the effective measurement noise, and individual beams can deviate substantially from the pre-built map. The lab therefore serves as a challenging environment for evaluating the filter’s robustness against inconsistent sensor measurements.

A second aspect is the sensitivity of the resulting trajectory to small changes in the initial pose. For this purpose, one undisturbed and one disturbed lab recording were evaluated repeatedly offline. This separates the effect of the difficult environment during normal operation from the filter’s ability to return to a consistent trajectory after stronger disturbances.

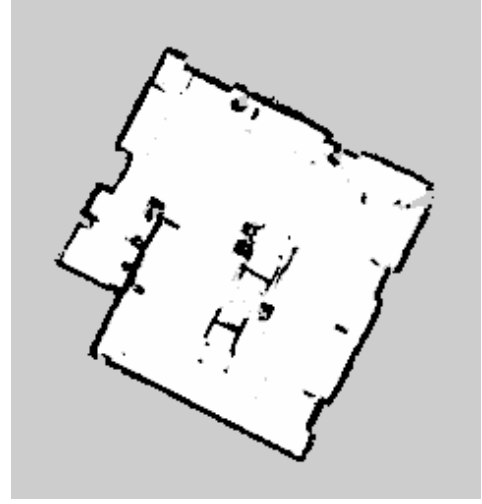


Fig. 6. Reference map of the lab environment with tables and chairs.

1) *Repeatability Experiment Setup:* A single EKF run provides only limited evidence about the filter’s robustness. Therefore, both lab recordings were evaluated offline 20 times. Only the initial pose was changed between runs, with offsets of at most 1 cm in position and 0.5° in orientation. Kidnapping recovery was disabled so that an incorrect local estimate could not be corrected through global relocalization.

This experiment does not investigate random behaviour under identical conditions. Instead, it shows how small differences in the initial state can affect the long-term state estimate.

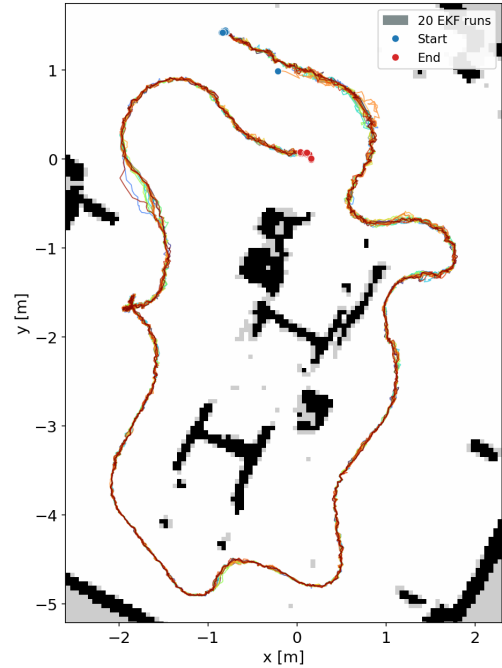


Fig. 7. Twenty EKF trajectories from the slow lab run between 70 and 341 s. Blue points mark the beginning and red points mark the end of the displayed interval.

2) *Undisturbed Slow Run*: Figure 7 shows a slow run without an intentionally introduced disturbance. Despite the slightly different initial poses, the 20 estimates follow nearly the same trajectory over most of the run. Visible deviations occur mainly in individual cluttered regions of the lab but remain limited. Under calm conditions, the EKF therefore produces largely repeatable results even in the challenging lab environment.

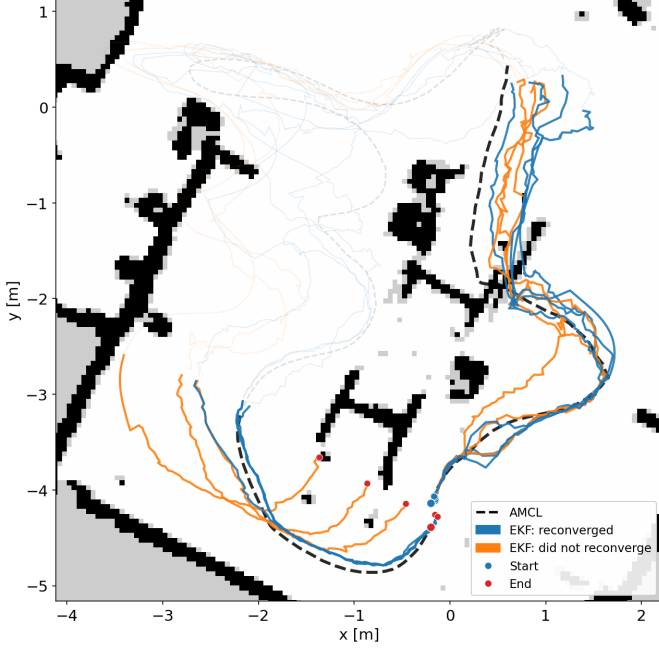


Fig. 8. Seven representative EKF trajectories during a disturbed lab run from 122 to 215 s. The interval from 150 to 200 s is shown transparently for readability. Blue denotes runs that reconverge, while orange denotes runs that remain divergent. The dashed AMCL path serves as a reference.

3) *Disturbed Run*: Stronger disturbances produce a different result. The trajectories in Fig. 8 initially remain close together but develop substantially different estimates after the disturbance.

Seven representative trajectories are shown from the total of 20 runs to keep the plot readable. Four estimates shown in blue reconverge and subsequently remain close to the AMCL reference path. Three estimates shown in orange do not return to a consistent path and remain incorrectly localized until the end of the displayed interval. AMCL also exhibits visible deviations during the disturbance but stabilizes afterwards. The selected runs provide a qualitative illustration and do not represent a success rate.

C. Kidnapping Recovery

The practical recovery behaviour was evaluated in the elevator room and the lab. During each run, the robot was manually lifted and placed at another location three times. The resulting pose estimate was assessed visually by comparing the LIDAR scan with the map. No external ground truth was used. Consequently, this experiment evaluates whether the system

returns to a plausible, map-aligned pose rather than its absolute localisation error.

In Figs. 9 and 10, the vertical lines mark the manual kidnappings, the blue regions show individual recovery attempts, and the horizontal lines indicate the mean number of valid beams during each tracking phase. Green regions denote visually correct tracking. The red region in Fig. 9 denotes a visually incorrect pose that was initially accepted as tracking. The dashed line is the lost threshold of 7.5 valid beams out of 30.

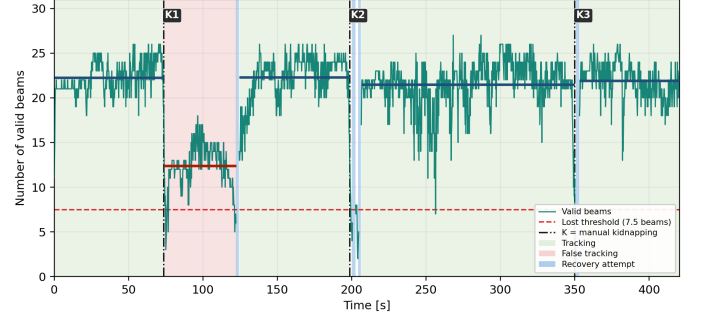


Fig. 9. Valid beams and recovery attempts during three manual kidnappings in the elevator room.

1) *Elevator Results*: During correctly aligned tracking in the elevator room, the filter maintained an average of approximately 21.9 valid beams. After the first kidnapping at 73.6 s, however, the filter resumed tracking at an incorrect pose. This failure remained undetected for approximately 48.4 s because the incorrect pose still produced an average of 12.4 valid beams, above the configured lost threshold. Once the agreement deteriorated further, the filter declared itself lost and the first recovery attempt restored correct tracking.

After the second kidnapping, the first recovery hypothesis was also incorrect. Recovery had already started while the robot was still being manually repositioned, so the first hypothesis could not match its final position. The system detected this mismatch after approximately 2 s, and the second attempt recovered the correct pose. The first recovery attempt after the third kidnapping was successful.

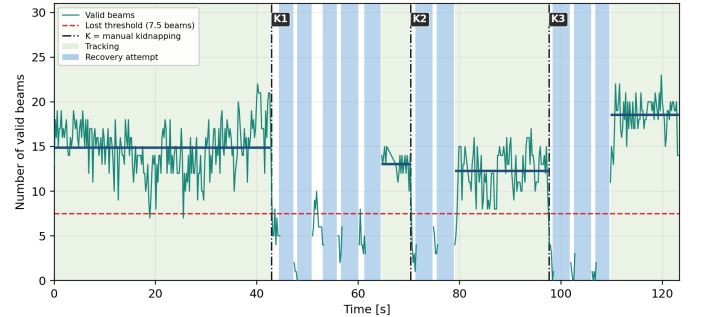


Fig. 10. Valid beams and recovery attempts during three manual kidnappings in the lab.

2) *Lab Results*: The correctly aligned lab phases contained only approximately 14.8 valid beams on average, compared

with 21.9 in the elevator room. This difference is consistent with the cluttered environment discussed in the Lab Room robustness experiment. Furniture and thin obstacles appear inconsistently in the LIDAR scan depending on the robot pose, resulting in less reliable sensor-map agreement. The three kidnappings required five, two, and three recovery attempts, respectively, before a visually correct pose was restored.

The results demonstrate that lost detection based on a fixed valid-beam threshold is strongly environment-dependent. Raising the threshold towards 12 beams would detect the incorrect elevator pose earlier, but would also approach the normal tracking level observed in the lab and could therefore cause false lost detections. Reliable recovery across structurally different environments consequently requires either environment-dependent thresholds or an additional localisation-quality criterion.

D. Hallway — Parameter Sensitivity

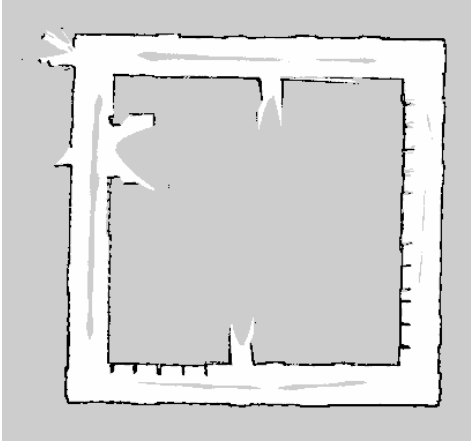


Fig. 11. Reference map of the hallway environment.

The hallway is a long, narrow corridor whose two parallel walls produce a largely uniform LIDAR signature with little variation along the direction of travel. This geometric ambiguity makes the environment particularly sensitive to the ratio between $\mathbf{Q}_k$ and $\mathbf{R}_k$, which determines how much the filter trusts odometry relative to sensor measurements.

The starting point for this experiment was the baseline parameter set (EKF 1 $\times$). With these settings the robot came to a standstill partway through the corridor. Because the two parallel walls look identical at every position along the hallway, the ray-casting model returns nearly the same expected measurement regardless of how far the robot has travelled. The innovation $\mathbf{y} = \mathbf{z} - h(\hat{\mathbf{x}})$ therefore stays close to zero, and the filter, placing high confidence in the sensor, concludes that no forward motion has occurred. Effectively, a small $\mathbf{R}_k$ made the EKF override the odometry signal with an uninformative measurement, causing localisation to stall.

To resolve this, we shifted trust from the sensor back towards odometry by applying the same $k\times$ scheme as in the elevator-room sweep. The EKF 2 $\times$ configuration already advanced further along the corridor before stalling, and EKF 3 $\times$

resolved the problem entirely. As visible in Fig. 12, the estimated trajectory then remained aligned with the AMCL ground truth throughout the full run, achieving an RMSE that even falls below the pure odometry error (Tab. II).

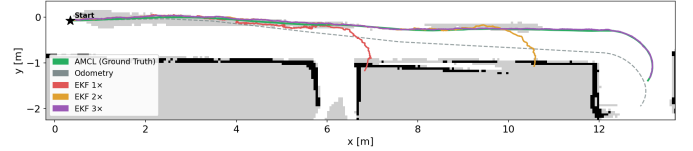


Fig. 12. Trajectory comparison in the hallway. The EKF 3 $\times$ configuration tracks the ground truth closely across the full corridor.

TABLE II
LOCALIZATION RMSE (M) FOR THE HALLWAY RUN.

Configuration	$(\sigma_{QD}, \sigma_{Q\theta}, \sigma_R)$	RMSE
Odometry	—	0.334
EKF 1 $\times$	(0.0500, 0.090, 0.06)	3.676
EKF 2 $\times$	(0.0250, 0.045, 0.12)	1.497
EKF 3 $\times$	(0.0167, 0.030, 0.18)	0.077

V. CONCLUSION AND OUTLOOK

The experiments across three structurally different environments demonstrate that the EKF with a ray-casting sensor model generalises well to varying scenarios. In all cases the filter converged to accurate estimates and tracked the robot reliably over full runs, including under external disturbances, and the kidnapping-recovery routine reliably re-localised the robot after manual displacement. The core algorithm, including the ray-casting model, numerical Jacobian, innovation gate, and kidnapping recovery, required no structural changes across scenarios. The only adjustment between environments was a retuning of the noise parameters $\mathbf{Q}_k$ and $\mathbf{R}_k$, and the hallway run illustrates how strong this dependency is: with the baseline $\mathbf{R}_k$ that performed well in the cluttered lab, the filter stalled completely and became unusable, while a shift of the $\mathbf{Q}_k/\mathbf{R}_k$ ratio towards odometry trust brought the RMSE below the pure-odometry error. Matching the parameter choice to the system and the environment is therefore essential, and the appropriate trade-off depends on how much corrective information the LIDAR signature can actually provide.

A notable limitation surfaces in symmetric environments such as the hallway, where the four corners are geometrically nearly identical. When kidnapping recovery scores hypotheses purely on local sensor-map agreement, all four corners yield the same score, and the filter may resume tracking from the wrong position without detecting the error. A promising extension would be to add a distance penalty to the hypothesis score based on the last known pose before the kidnapping event, under the assumption that the robot has not been moved arbitrarily far. This would break the symmetry between equally scoring locations and guide recovery towards the most plausible hypothesis.

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- [1] A. A. Housein, G. Xingyu, W. Li, and Y. Huang, "Extended kalman filter sensor fusion in practice for mobile robot localization," *International Journal of Advanced Computer Science and Applications*, vol. 13, no. 2, 2022. [Online]. Available: <http://dx.doi.org/10.14569/IJACSA.2022.0130204>